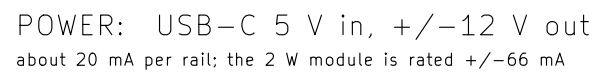


...built here as 2.2 nF always fitted, plus 100 nF and 470 nF switched in by SW1. See the notes at the right of the sheet.



one 0.1 uF at every supply pin, right at the package



Each op-amp is an integrator with the terms of one derivative summed at its inverting input, so the weight of every term is simply $1 \text{ MEG} / R$:

R2 = R1 = 100k $\rightarrow$ 1M/100k = 10.00 = s (and the -x damping)
R3 = 35.7k $\rightarrow$ 1M/35.7k = 28.01 = r
R5 = 1M $\rightarrow$ 1M/1M = 1.00 = the -y term
R7 = 374k $\rightarrow$ 1M/374k = 2.674 = b (8/3 = 2.667, 0.3 % high)
R4 = R6 = 10k $\rightarrow$ 1M/10k = 100 cancels the multipliers' /100

Voltages carry the dimensionless variables at 0.1 V per unit, so x swings +/-2 V, y +/-2.7 V and z runs 0 to 4.8 V. An MPV634 forms A*B/10, so the product of two 0.1 V/unit signals comes out 100x too small -- which is exactly what the 10k weighting resistors R4 and R6 put back. The minus signs on both products are free: U3 and U4 simply have their differential inputs swapped.

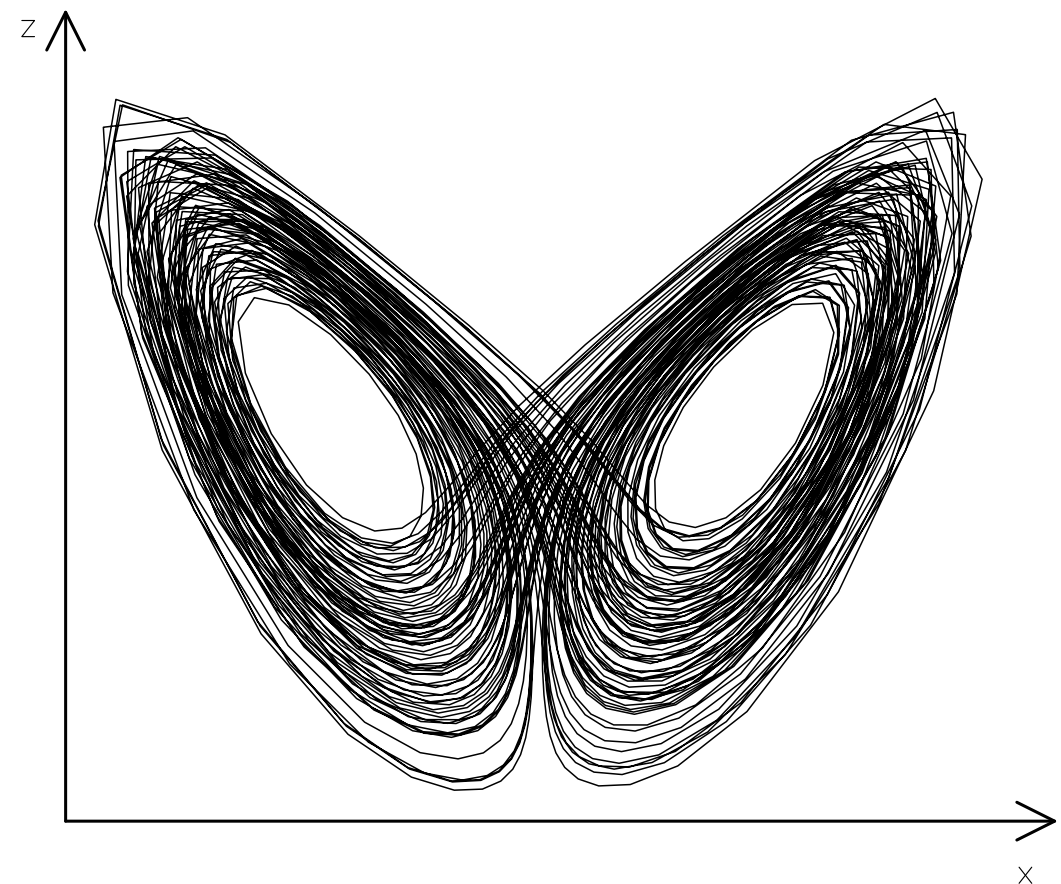
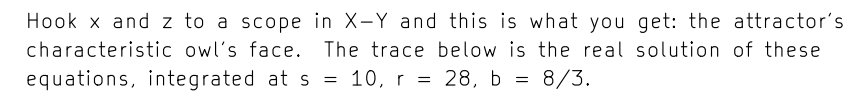
The time scale is $\tau = 1 \text{ MEG} \times C$, and C is switchable:

SW1-1,2,3 off	SW1-4,5,6 off	C = 2.2 nF	fast!	~3 ms per lobe
SW1-1,2,3 ON	SW1-4,5,6 off	C = 102 nF	nice!	~0.15 s per lobe
SW1-1,2,3 off	SW1-4,5,6 ON	C = 472 nF	slow!	~0.7 s per lobe

Paul's original ran on a bench ± 15 V supply and used 2000 pF, 0.1 μ F and 0.47 μ F. Here the three values are built up in parallel because no supplier stocks a 3-pole 3-position switch; 2.2 nF is always fitted.

MOUNTING

Four M3 holes, one per board corner. No electrical function, but with three BNCs along one edge the cables lever on the board, so it wants standoffs rather than a bare bench.



Open source -- https://github.com/gallicchio/ai-unleashes_chaos
 Every coefficient is 1 MEG / R; signals are 0.1 V per unit
 $s = 10, r = 28, b = 8/3$
 $dx/dt = s(y-x) \quad dy/dt = rx - y - xz \quad dz/dt = xy - bz$
Circuit by Paul Horowitz | PCB by Jason Gallicchio and Claude
 Sheet: /
 File: lorenz-4layer.kicad_sch

Title: AI Unleashes Chaos -- Lorenz Attractor

Size: A2	Date: 2026-09-15
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Rev: A